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殷家兴 自动化

数电⑩

6-5

解:

~~RC~~ 假设 $R_1 + R_5 \gg R$, 则

$$T_2 = RC \ln \frac{2U_{OL} - U_{TH} - U_{OH}}{U_{OL} - U_{TH}}$$

$$T_1 = RC \ln \frac{2U_{OH} - U_{TH} - U_{OL}}{U_{OH} - U_{TH}}$$

~~$$T_2 = RC \ln \frac{U_{OL} - [U_{TH} + (U_{OH} - U_{OL})]}{U_{OL} - U_{TH}}$$~~

取 $U_{OH} = 3.6V$ $U_{OL} = 0.3V$ $U_{TH} = 1.4V$

则 $T_2 = 100 \times 0.01 \times 10^{-6} \times \ln \frac{2 \times 0.3 - 1.4 - 3.6}{0.3 - 1.4} s$

$$= 2 \ln 2 \times 10^{-6} s = 1.39 \times 10^{-6} s$$

~~$$T_2 = 100 \times 0.01 \times 10^{-6} \times \ln \frac{0.3 - [1.4 + (3.6 - 0.3)]}{0.3 - 1.4} s$$~~

$$T_1 = 100 \times 0.01 \times 10^{-6} \times \ln \frac{2 \times 3.6 - 1.4 - 0.3}{3.6 - 1.4} s$$

$$= \ln \frac{5}{2} \times 10^{-6} s = 0.92 \times 10^{-6} s$$

$$T = T_1 + T_2 = 2.31 \times 10^{-6} s$$

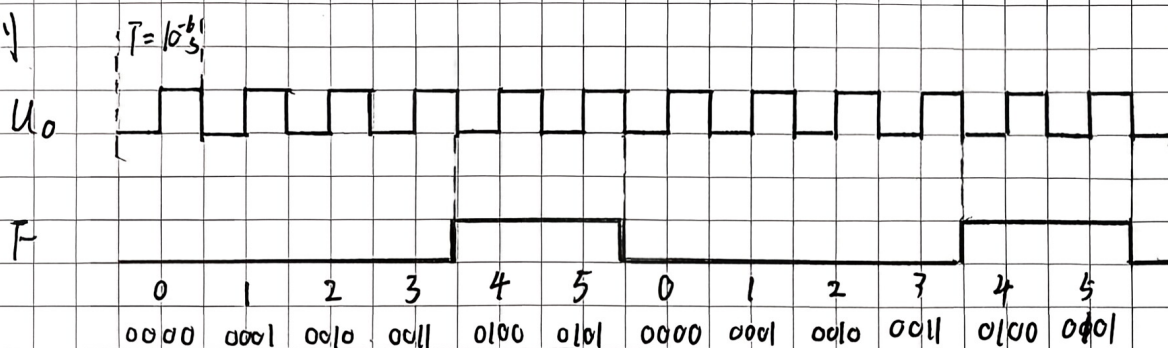
$$f = \frac{1}{T} = 4.33 \times 10^5 \text{ Hz}$$

6-6 解: $f = 1 \text{ MHz}$

$$T = \frac{1}{f} = 1 \times 10^{-6} s$$

此时 74LS90 CP_2 与 Q_A 相连, ~~构成 8421 BCD 计数器~~当 Q_B 、 Q_C 均为 1 时清零, 为模 6 计数器
同时

则



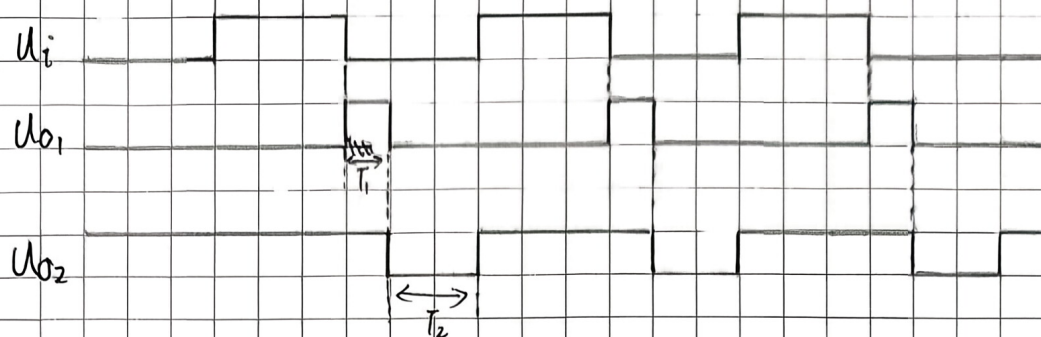
F 的周期 $T_F = 6T = 6 \times 10^{-6} s$

$$f_F = \frac{1}{T_F} = \frac{1}{6} \text{ MHz} \approx 0.167 \text{ MHz}$$

6-7 解:

$$T_1 = 0.7 R_1 C_1 = 1.001 \times 10^{-3} s \approx 1 ms$$

$$T_2 = 0.7 R_2 C_2 = 2.002 \times 10^{-3} s \approx 2 ms$$



6-8 解:

